

**California State University, San Bernardino**

**CSE 5210 – Digital System Design**

**Fall 2026**

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**Problem Set 1**

Due date: Monday 9/7/2024 11:59 PM

**Instruction:**

Please submit a PDF file containing your answers to these questions. You can either type out your responses or take clear and readable pictures of your handwritten answers. In the latter case, ensure that the images are saved in a Word file and then converted to PDF format for submission (only submit 1 file containing the answer to all questions). Remember to include detailed explanations, including the entire process and calculations, not just the final answers, to ensure you receive full credit.

**Problem 1**

Design a combinational circuit with four inputs and two outputs.

Inputs:

- A,B,C: represent a 3-bit user ID.
- D: represents an authorization flag (1 = authorized, 0 = not authorized).

Outputs:

- X: should be 1 if the user ID corresponds to an even number and the authorization flag is 1.

- Y: should be 1 if the user ID is greater than or equal to 5, regardless of the authorization flag.

Tasks:

1. Construct the truth table for the circuit.
2. Draw the K-map for each output.
3. Derive the simplified Boolean equations for X and Y.

## Problem 2

Design a circuit with four inputs (A,B,C,D) and one output F.

- The output  $F=1$  whenever the input is between 9 and 12 (inclusive).
- Use a 4-to-16 decoder and a 2-to-1 multiplexer in your design, while minimizing the use of additional logic gates.

Tasks:

1. Write the truth table.
2. Clearly describe your reasoning for using the decoder and the multiplexer in specific parts of the design, and provide a detailed explanation of all connections.
3. Draw the final circuit.